Lab 5.1 First Steps with Mininet: Topologies, CLI, and Traffic Capture

Task 1

```
mininet-wifi> dump
<Host h1: h1-eth0:10.0.0.1 pid=5648>
<Host h2: h2-eth0:10.0.0.2 pid=5650>
<Host h3: h3-eth0:10.0.0.3 pid=5652>
<OVSSwitch s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None,s1-eth3:N
one pid=5657>
<OVSController c0: 127.0.0.1:6653 pid=5641>
mininet-wifi> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
```

Task 2

```
mininet-wifi> iperf
*** Iperf: testing TCP bandwidth between h1 and h3
*** Results: ['16.4 Gbits/sec', '16.4 Gbits/sec']
mininet-wifi> h1 ping -c 5 h3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=1.22 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=0.195 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=0.039 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=64 time=0.038 ms
64 bytes from 10.0.0.3: icmp_seq=5 ttl=64 time=0.039 ms

--- 10.0.0.3 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4083ms
rtt min/avg/max/mdev = 0.038/0.306/1.219/0.460 ms
mininet-wifi>
```

Task 3

```
mininet-wifi> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s2-eth1
h3 h3-eth0:s3-eth1
h4 h4-eth0:s4-eth1
s1 lo:  s1-eth1:h1-eth0 s1-eth2:s2-eth2
s2 lo:  s2-eth1:h2-eth0 s2-eth2:s1-eth2 s2-eth3:s3-eth2
s3 lo:  s3-eth1:h3-eth0 s3-eth2:s2-eth3 s3-eth3:s4-eth2
s4 lo:  s4-eth1:h4-eth0 s4-eth2:s3-eth3
c0
```

```
mininet-wifi> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4
h2 -> h1 h3 h4
h3 -> h1 h2 h4
h4 -> h1 h2 h3
*** Results: 0% dropped (12/12 received)
mininet-wifi> h1 ping -c 5 h4
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.
64 bytes from 10.0.0.4: icmp_seq=1 ttl=64 time=0.402 ms
64 bytes from 10.0.0.4: icmp_seq=2 ttl=64 time=0.050 ms
64 bytes from 10.0.0.4: icmp_seq=3 ttl=64 time=0.054 ms
64 bytes from 10.0.0.4: icmp_seq=4 ttl=64 time=0.133 ms
64 bytes from 10.0.0.4: icmp_seq=5 ttl=64 time=0.049 ms

--- 10.0.0.4 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4127ms
rtt min/avg/max/mdev = 0.049/0.137/0.402/0.135 ms
mininet-wifi>
```

Answer: "Linear" configuration where four switches are connected in a sequence with a single host, but "Single-3" only has 1 switch and also connects every host directly to the switches.

Task 4

```
osboxes@osboxes:~/Desktop$ sudo mn --topo tree,depth=2,fanout=2
```

```
mininet-wifi> net
h1 h1-eth0:s2-eth1
h2 h2-eth0:s2-eth2
h3 h3-eth0:s3-eth1
h4 h4-eth0:s3-eth2
s1 lo:  s1-eth1:s2-eth3 s1-eth2:s3-eth3
s2 lo:  s2-eth1:h1-eth0 s2-eth2:h2-eth0 s2-eth3:s1-eth1
s3 lo:  s3-eth1:h3-eth0 s3-eth2:h4-eth0 s3-eth3:s1-eth2
c0
```

Actualresults: hosts = 4 (h1–h4), switches = 3 (s1–s3)
Formula: hosts = $2^2 = 4$, switches = $(2^2 - 1)/(2 - 1) = 3$
Result: Matches the formula

```
osboxes@osboxes:~/Desktop$ sudo mn --topo tree,depth=3,fanout=2
```

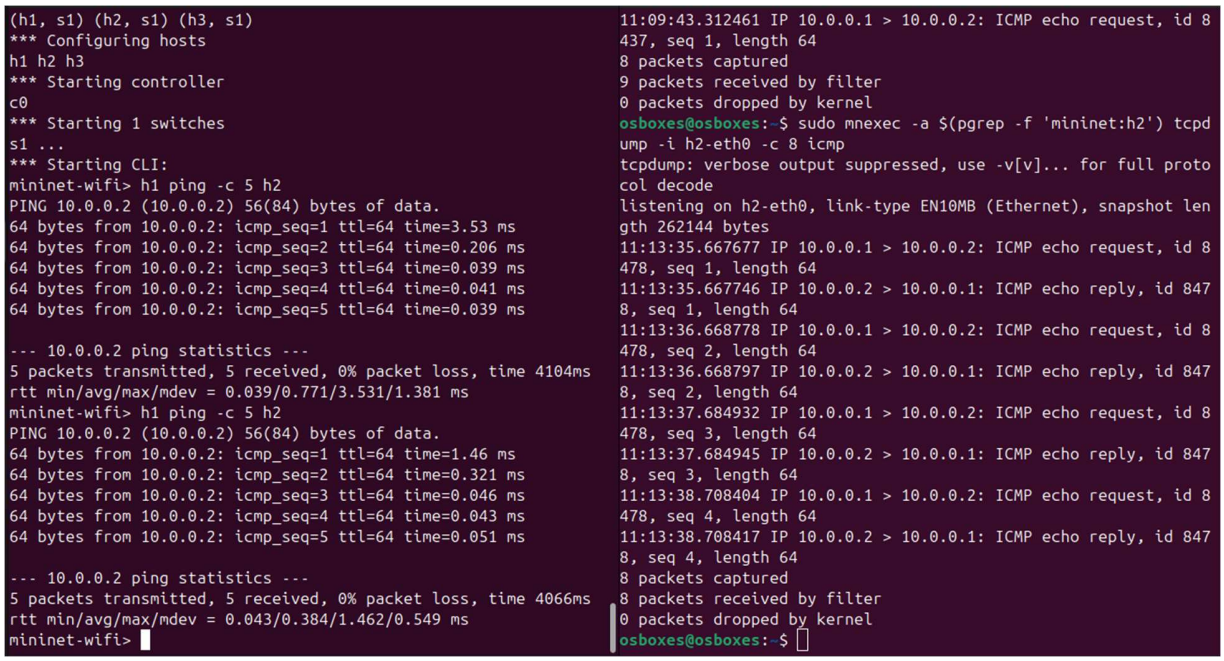
```
mininet-wifi> net
h1 h1-eth0:s3-eth1
h2 h2-eth0:s3-eth2
h3 h3-eth0:s4-eth1
h4 h4-eth0:s4-eth2
h5 h5-eth0:s6-eth1
h6 h6-eth0:s6-eth2
h7 h7-eth0:s7-eth1
h8 h8-eth0:s7-eth2
s1 lo:  s1-eth1:s2-eth3 s1-eth2:s5-eth3
s2 lo:  s2-eth1:s3-eth3 s2-eth2:s4-eth3 s2-eth3:s1-eth1
s3 lo:  s3-eth1:h1-eth0 s3-eth2:h2-eth0 s3-eth3:s2-eth1
s4 lo:  s4-eth1:h3-eth0 s4-eth2:h4-eth0 s4-eth3:s2-eth2
s5 lo:  s5-eth1:s6-eth3 s5-eth2:s7-eth3 s5-eth3:s1-eth2
s6 lo:  s6-eth1:h5-eth0 s6-eth2:h6-eth0 s6-eth3:s5-eth1
s7 lo:  s7-eth1:h7-eth0 s7-eth2:h8-eth0 s7-eth3:s5-eth2
```

Predicted values: hosts = $2^3 =$ **8**, switches = $(2^3 - 1)/ (2 - 1) =$ **7**
Actual network results: hosts = 8 (h1–h8), switches = 7 (s1–s7)
Result: Matches the prediction

Task 5

```
mininet-wifi> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet-wifi> link h1 s1 down
mininet-wifi> pingall
*** Ping: testing ping reachability
h1 -> X X
h2 -> X h3
h3 -> X h2
*** Results: 66% dropped (2/6 received)
mininet-wifi> link h1 s1 up
mininet-wifi> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet-wifi>
```

Task 6



Answer: The fact that tcpdump successfully captures actual ARP and ICMP frames on the h2-eth0 interface demonstrates that Mininet utilizes the actual Linux network stack and transmits real packets (emulation), rather than merely simulating events using models like those found in simulators such as NS-3 or OMNeT++.

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Reflection

1. Compare the linear and tree topologies in terms of how many switch hops separate the two farthest-apart hosts as the network grows larger. Which topology scales better, and why?
   Answer: In a linear topology, the hop distance between the two most distant hosts grows linearly with the number of switches (n), whereas in a tree topology, it grows logarithmically (approximately 2 × depth); consequently, the tree topology scales better.

2. Why might the average ping RTT you measured in Part C (linear,4, host to host across the whole chain) be higher than the RTT you measured in Part A (single,3)?
   Answer: In a linear-4 topology, the path from h1 to h4 traverses four switches, with each hop incurring OVS processing and packet copying overhead, whereas in a single-3 topology, it passes through only one switch. If your actual measurements show the linear topology yielding a lower RTT (0.137 vs. 0.306 ms), attribute this to the initial ping—which involves waiting for ARP resolution and flow installation—and to CPU jitter, as measurements at the sub-millisecond level are prone to high noise.

3. Based on today's hands-on experience, describe one scenario where you would choose Mininet-style emulation for a project, and one scenario where a pure simulator like NS-3 or OMNeT++ would be the better choice instead?
   Answer: Use Mininet when testing code or an actual SDN controller with real packets. Use NS-3 or OMNeT++ when simulating very large-scale networks or requiring highly reproducible results that are independent of the host machine's CPU.